Yasaman Jafari

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Education

University of California San Diego, Ph.D. in Computer Science — Oct. 2021 – Present

- **GPA:** 4.0/4.0
- **Advised by:** Dr. Taylor Berg-Kirkpatrick
- **M.Sc. Degree in Computer Science** acquired Dec. 2024
- **Advanced to Candidacy** Dec. 2025

University of Tehran, B.Sc. in Computer Engineering — Sept. 2016 – Feb. 2021

- **GPA:** 19.14/20.0 (4.0/4.0)
- **Advised by:** Dr. Behnam Bahrak
- **B.Sc. Thesis:** Investigating the effects of Goodreads' challenges on individuals' reading habits

Research Experience

Graduate Research Assistant, University of California San Diego – California, United States (Advisor: Dr. Taylor Berg-Kirkpatrick) — Jun. 2023 – Present

- **Scalable Corpus Reasoning via "Document Souping" in State Space Models:** Designed a modular procedure that enables parallel processing and caching of document representations, eliminating the need for costly re-encoding of the entire corpus for each query.
- **GENIE - Climate Co-Scientist:** Building an agentic AI framework and novel benchmarks to accelerate weather and climate science research. (In Progress)
- **RL-based Discrete Prompt Optimization:** Developed multi-objective optimization techniques for RL-based prompt optimization, improving balance across competing rewards in NLP tasks like style transfer and translation.

Graduate Research Assistant, University of California San Diego – California, United States (Advisor: Dr. Babak Salimi) — Oct. 2021 – Jun. 2023

- Repaired datasets using optimal transport to enforce conditional independence constraints and improve robustness
- Quantified subgroup heterogeneity and intersectional effects to mitigate performance disparities and improve calibration

Undergraduate (Volunteer) Research Assistant, University of Tehran Science and Technology Park – Tehran, Iran (Advisor: Dr. Behnam Bahrak) — Oct. 2019 – Feb. 2021

- Analyzed GoodReads challenges and the potential effects of publicly announcing goals

Research Papers

Black-Box Forensics for Conversational LLM Agents — 2026
I. White, **Y. Jafari**, T. Berg-Kirkpatrick
[Under Submission]

Studying the Soupability of Documents in State Space Models — 2025
Y. Jafari, Z. Wang, L. Bergen, T. Berg-Kirkpatrick
[COLM 2026]

Zephyrus: An Agentic Framework for Weather Science — 2025
S. Varambally, M. Fisher, J. Thakker, Y. Chen, Z. Xia, **Y. Jafari**, R. Niu, M. Jain, V. V. Manivannan, Z. Novack, L. Han, S. Eranky, S. R. Cachay, T. Berg-Kirkpatrick, D. Watson-Parris, Y. Ma, R. Yu
[ICLR 2026]

Zephyrus: An Agentic Framework for Weather Science — 2025
S. Varambally, M. Fisher, J. Thakker, Y. Chen, Z. Xia, R. Niu, **Y. Jafari**, V. V. Manivannan, Z. Novack, L. Han, S. Eranky, S. R. Cachay, T. Berg-Kirkpatrick, D. Watson-Parris, Y. Ma, R. Yu
[NeurIPS 2025 Workshop on Scaling Environments for Agents (SEA)]

Aquilon: Towards Building Multimodal Weather LLMs 2025
S. Varambally, V. Manivannan, **Y. Jafari**, L. Han, Z. Novack, Z. Xia, S. Rühling Cachay, S. Eranky, R. Niu, T. Berg-Kirkpatrick, D. Watson-Parris, Y. Ma, R. Yu
[ICML 2025 Workshop on Assessing World Models]

ClimaQA: An Automated Evaluation Framework for Climate Foundation Models 2024
V. Manivannan, **Y. Jafari**, S. Eranky, S. Ho, R. Yu, D. Watson-Parris, Y. Ma, L. Bergen, T. Berg-Kirkpatrick
[ICLR 2025] arXiv:2410.16701

MORL-Prompt: An Empirical Analysis of Multi-Objective Reinforcement Learning for Discrete Prompt Optimization 2024
Y. Jafari, D. Mekala, R. Yu, T. Berg-Kirkpatrick
[EMNLP Findings 2024] arXiv:2402.11711

Investigating the effects of Goodreads challenges on individuals reading habits 2020
Y. Jafari, N. Sabri, B. Bahrak
[Pre-print] arXiv:2012.03932

Work/Internship Experience

Siri SW Intern, Apple Inc. – Cupertino, United States — Jun. - Sept. 2026
- Research on State Space Models for Agentic models

Software Development Engineer 2, Populus Group (On Assignment for Microsoft Research - Continuation of Summer 2025 MSR Internship work) – Part-time, United States (Remote) — Feb. - Jun. 2026
- AI for Science and Agentic Frameworks

Research Intern, Microsoft Research – Redmond, United States — Jun. – Sept. 2025
- Developed an agentic framework for power grid planning and risk management from hurricanes and other extreme weather events

Data Science Intern (KYC - Adverse Media), Moody's Analytics (Moody's) – New York, United States (Remote) — Jun. - Aug. 2023
- Created evaluation data for an existing model used in Adverse Media Highlighting
- Evaluated the model and conducted a thorough analysis of its performance, identifying areas of failure, and proposing various options for improving the model

Data Analysis and Modeling Intern, Moody's Analytics (Moody's) – United States (Remote) — Jun. - Aug. 2022
- Mitigated the run-time bottlenecks of working with large data using Spark
- Converted estimation codes from R to Python for efficiency purposes

Software Engineering Intern, Supertext AG – Zurich, Switzerland — Feb. - Jul. 2020
- Implemented various front-end and back-end features such as Instant Translation and SSO authentication

Teaching Experience

Graduate Teaching Assistant, University of California San Diego – California, United States
- CSE 251A: Machine Learning: Learning Algorithms, Dr. Taylor Berg-Kirkpatrick — Jan. – Mar. 2026
- CSE 158: Web Mining and Recommender Systems, Dr. Julian McAuley — Sept. – Dec. 2023
- CSE 256: Statistical Natural Language Processing, Dr. Ndapa Nakashole — Mar. – Jun. 2023
- CSE 250A: Probabilistic Reason and Learning Course, Dr. Taylor Berg-Kirkpatrick — Sept. - Dec. 2022

Honors and Awards

- Google CS Research Mentorship Program (CSRMP) Scholar — May. 2023
- Fellowship Award for Graduate Study at UC San Diego — Oct. 2021 - Jun. 2022
- Finalist for the Best Undergraduate Project Award — Feb. 2021
- 3rd rank (among 88 students) - FOE Top Students Award — Jun. 2018
- 1st rank (among 88 students) - FOE Top Students Award — Jun. 2017

Services

Conference Reviewer: COLM 2026, ICLR 2026, NeurIPS 2025, ICML 2025, NeurIPS 2024, ICML 2024, ICLR 2024, NeurIPS 2023, ICML 2023, FAccT 2023
UCSD Graduate Women in Computing (GradWIC) Mentor Served as a mentor in the GradWIC mentorship program for incoming graduate students from underrepresented backgrounds (Academic year 2023-2024)
Conference Volunteer Student ICDE 2023

Skills

Programming Languages: Python, C++, C, C#, Java, JavaScript, Ruby

Languages: Persian (Native Language), English (Professional Working Proficiency – TOEFL 119/120), German (Intermediate – B1)